



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Certified Lean Six Sigma Black Belt (CLSSBB) Training

TGS Ref No: TGS-2024051900

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 4

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1	1 June 2026	Initial release - CLSSBB lesson plan.	Dr. Alfred Ang
2	23 September 2026	Major revision: rebuilt as a 5-day Black Belt progression from the Green Belt course (CLSSGB). DMAIC is recapped IN DEPTH (a Black Belt must be able to mentor Green Belts through these tools) and then extended with Black-Belt-only content the Green Belt course does not teach: attribute MSA and Kappa analysis, nested and destructive gauge studies, non-normal capability and Cpk versus Ppk, multiple regression with model diagnostics and VIF, ANOVA with interaction effects, non-parametric tests, multi-vari studies, full and fractional factorial Design of Experiments with confounding and resolution, response surface methodology, Taguchi robust design, CUSUM, EWMA, short-run and multivariate SPC, change leadership and resistance management, financial benefits validation, and mentoring Green Belts. Expanded to 34 hands-on labs across five days. THE CAPSTONE PROJECT RUNS FROM DAY 1 - each learner selects a real workplace process in Lab 1 and every subsequent lab adds one artifact to it; Day 5 consolidates the artifacts into an A3	Dr. Alfred Ang

		storyboard and a steering committee presentation.	
3	23 September 2026	Corrected the assessment model to match the registered WSQ instrument: the assessment is the Written Assessment (WA) Short-Answer Questions plus the Case Study (CS), both open book. The capstone project and steering committee presentation are retained as an ADDITIONAL course deliverable on top of the WSQ assessment, not as a replacement for it. Added the online SIPOC tool (alfredang.github.io/sipoc) to the toolkit, Lab 6 and the Define slides.	Dr. Alfred Ang
4	23 September 2026	Hands-on lab data release. Every lab now has its OWN folder under labs/ containing the worksheet and an Excel workbook: 20 labs carry simulated process datasets and 14 carry structured templates. All data describes one running process (the Meridian Medical Devices seal-weld line), so figures reconcile across labs — baseline Ppk 0.62 in Lab 14, the proven Xs in Labs 18-21, the DOE optimum in Labs 23-25, improved Ppk 1.40 in Lab 27 and the SGD 487,000 benefit in Lab 30 against the SGD 812,000 COPQ chartered in Lab 1. Every workbook opens on a Data Dictionary sheet. The Lesson Plan adds a Lab Data Files and Materials table so the trainer can set the room up before the session.	Dr. Alfred Ang

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Course Information

Course Title	Certified Lean Six Sigma Black Belt (CLSSBB) Training
WSQ Course Reference	TGS-2024051900
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	5 days · 8 training hours per day (40 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on Lean Six Sigma labs applied to each learner's OWN workplace capstone project (Meridian Medical Devices scenario provided as a fallback)
TSC Alignment	Quality Process Control (ELE-QUA-5006-1.1)
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Lead a Lean Six Sigma project portfolio — select, charter and govern projects aligned to business strategy (A1, A2).
- LO2: Validate measurement systems for continuous AND attribute data using Gage R&R and Kappa analysis (A4).
- LO3: Analyse process data using multiple regression, ANOVA, non-parametric tests and multi-vari studies (A3).
- LO4: Design, run and interpret experiments using full and fractional factorial DOE and response surface methods (A3, K2).
- LO5: Select, de-risk and pilot improvements, and validate financial benefits with Finance sign-off (A5).
- LO6: Sustain the gain using advanced SPC — CUSUM, EWMA, short-run and multivariate control charts (A5, K1).
- LO7: Lead change and mentor Green Belts through resistance, stakeholder management and tollgate governance (A1, K1).
- LO8: Deliver a capstone improvement project and present it to a steering committee (A1-A5, K1, K2).

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 2 questions, 1 hour, open book (K1, K2).
- Case Study (CS) — an applied DMAIC case study, 90 minutes, open book (A1-A5).
- Capstone Project & Presentation — an ADDITIONAL course deliverable on top of the WSQ assessment: the consolidated DMAIC storyboard built across Labs 1-30, presented and defended to a steering committee on Day 5.
- Format: Open Book — course slides, Learner Guide and approved materials only.

- The WSQ assessment is conducted on Day 5: WA (SAQ) from 4:00 pm (1 hour), then the Case Study from 5:00 pm (90 minutes).
- The capstone presentation is delivered on Day 5 and is additional to the WSQ assessment.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — The Black Belt role, portfolio selection, capstone chartering and Define

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, ground rules and mandatory digital attendance (AM). Briefing on the capstone project: every learner selects a real workplace process today and builds it across all five days [slides 2–22]
10:00–11:15	75 min	FOUNDATIONS — the Black Belt role versus the Green Belt role; leading a portfolio; owning the advanced statistics; mentoring Green Belts; project portfolio selection criteria; strategic (Hoshin) alignment; Cost of Poor Quality at portfolio scale [slides 23–93]
11:15–11:30	15 min	Tea break
11:30–12:30	60 min	FOUNDATIONS — DMAIC recap at depth: the purpose, core tools, tollgate deliverable and common failure mode of each phase; $Y = f(X)$; sigma level, DPMO and the 1.5 sigma shift; tollgate governance
12:30–13:30	60 min	Lunch break
13:30–15:15	105 min	Hands-on: Lab 1: Select and Charter Your Capstone Project; Lab 2: Portfolio Selection, Strategy Alignment and the Black Belt Role; Lab 3: DMAIC at Depth — Recap, $Y = f(X)$ and Baseline Sigma for Your Project [slides 94–105]
15:15–15:30	15 min	Tea break
15:30–17:00	90 min	DMAIC · DEFINE — VOC, affinity and Kano in depth; CTQ trees, operational definitions and specification limits; CTQ FLOWDOWN from business Y to controllable Xs; charter

		governance; stakeholder influence/support mapping; resistance diagnosis; ADKAR and Kotter change leadership [slides 106–179]
17:00–18:15	75 min	Hands-on: Lab 4: Voice of the Customer, Affinity and Kano — In Depth; Lab 5: CTQ Trees and CTQ Flowdown to Controllable Xs [slides 180–187]
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 2 — Measure — advanced measurement systems, attribute MSA and capability

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	Hands-on: Lab 6: SIPOC, Process Mapping and Scope Governance; Lab 7: Stakeholder Analysis, Resistance Management and Change Leadership [slides 188–196]
11:00–11:15	15 min	Tea break
11:15–12:30	75 min	DMAIC · MEASURE — process mapping, swimlanes and handoff analysis; value stream mapping with material and information flow; the timeline ladder and value-added ratio; takt time; the eight wastes (DOWNTIME); data types and operational definitions; the data collection plan [slides 197–290]
12:30–13:30	60 min	Lunch break
13:30–15:00	90 min	Hands-on: Lab 8: Value Stream Mapping, Takt Time and the Eight Wastes — In Depth; Lab 9: Data Types, Operational Definitions and the Data Collection Plan [slides 291–298]
15:00–15:15	15 min	Tea break
15:15–16:45	90 min	DMAIC · MEASURE — sampling schemes, sample size formulas and RATIONAL SUBGROUPING; continuous Gage R&R in depth (%study variation, ndc); ATTRIBUTE MSA and KAPPA analysis; nested and destructive

		gauge studies; yield, RTY and DPMO; non-normal capability, Box-Cox, Cpk versus Ppk
16:45–18:15	90 min	Hands-on: Lab 10: Sampling Strategy, Sample Size and Rational Subgrouping; Lab 11: Continuous Gage R&R — In Depth; Lab 12: Attribute MSA and Kappa Analysis (Black Belt Only) [slides 299–310]
18:15–18:30	15 min	Day 2 recap and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 3 — Analyze — multiple regression, ANOVA, non-parametric and multi-vari

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 2 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	Hands-on: Lab 13: Nested and Destructive Gauge Studies (Black Belt Only); Lab 14: Baseline Capability — Cp, Cpk, Pp, Ppk and Non-Normal Data [slides 311–319]
11:00–11:15	15 min	Tea break
11:15–12:30	75 min	DMAIC · ANALYZE — common versus special cause and tampering (the Deming funnel); run charts and the six non-random patterns; Pareto weighted by cost; stratification and boxplots; fishbone 5M+E; 5 Whys; multi-voting and the cause-and-effect matrix [slides 320–419]
12:30–13:30	60 min	Lunch break
13:30–14:45	75 min	Hands-on: Lab 15: Variation, Run Charts, Pareto and Stratification — In Depth; Lab 16: Fishbone, 5 Whys and Cause Prioritisation — In Depth [slides 420–427]
14:45–15:00	15 min	Tea break
15:00–16:45	105 min	DMAIC · ANALYZE — hypothesis testing in depth (alpha, beta, POWER, test selection logic, assumption checking); MULTIPLE REGRESSION and model diagnostics (adjusted R-squared, residual plots, multicollinearity and VIF); ANOVA and alpha inflation; two-way ANOVA and INTERACTION effects; non-parametric tests; MULTI-

16:45–18:15	90 min	VARI studies; chi-square Hands-on: Lab 17: Hypothesis Testing — In Depth, With Test Selection; Lab 18: Multiple Regression and Model Diagnostics (Black Belt Only); Lab 19: ANOVA, Interactions and Non-Parametric Tests (Black Belt Only) [slides 428–441]
18:15–18:30	15 min	Day 3 recap and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 4 — Improve and Control — DOE, RSM and advanced SPC

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 3 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	Hands-on: Lab 20: Multi-Vari Studies and Chi-Square Analysis (Black Belt Only); Lab 21: Analyze Tollgate — Proving Root Cause and Telling the Data Story [slides 442–451]
11:00–11:15	15 min	Tea break
11:15–12:30	75 min	DMAIC · IMPROVE — solution generation and benchmarking; Lean countermeasures (5S, poka-yoke, pull, JIT, Heijunka, Jidoka, standard work); the weighted solution selection matrix; FMEA, RPN, the severity override and re-scoring; DESIGN OF EXPERIMENTS: why OFAT fails, full factorial 2^k, randomisation, replication, blocking, main effects and interaction plots [slides 452–540]
12:30–13:30	60 min	Lunch break
13:30–15:00	90 min	Hands-on: Lab 22: Solution Generation, Selection and Lean Countermeasures — In Depth; Lab 23: Full Factorial Design of Experiments (Black Belt Only) [slides 541–549]
15:00–15:15	15 min	Tea break
15:15–16:45	90 min	DMAIC · IMPROVE and CONTROL — fractional factorial designs, confounding, alias structure and RESOLUTION III/IV/V; fold-over designs; response surface methodology, centre points, central composite designs; Taguchi robust

		design; SPC recap and chart selection; CUSUM, EWMA, short-run and MULTIVARIATE SPC; control plan governance; financial benefits validation [slides 566–639]
16:45–18:15	90 min	Hands-on: Lab 24: Fractional Factorial Designs, Confounding and Resolution (Black Belt Only); Lab 25: Response Surface Methodology and Robust Design (Black Belt Only); Lab 26: FMEA, Pilot Design and Implementation Planning — In Depth; Lab 27: Statistical Process Control and Chart Selection — In Depth [slides 550–643]
18:15–18:30	15 min	Day 4 recap and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 5 — Capstone project consolidation and steering committee presentation

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 4 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	Hands-on: Lab 28: CUSUM, EWMA and Short-Run SPC (Black Belt Only); Lab 29: Multivariate SPC and Control Plan Governance (Black Belt Only) [slides 644–653]
11:00–11:15	15 min	Tea break
11:15–11:45	30 min	Hands-on: Lab 30: Financial Benefits Validation, Handover and Project Closure [slides 654–659]
11:45–12:30	45 min	CAPSTONE — consolidating your artifacts from Labs 1-30; the A3 storyboard structure (Background, Current State, Goal, Root Cause, Countermeasures, Results, Follow-up); building the steering committee presentation; the two-minute rule and the appendix strategy [slides 660–697]
12:30–13:30	60 min	Lunch break
13:30–14:15	45 min	Hands-on: Lab 31: Consolidating Your Capstone — The A3 Storyboard; Lab 32: Building the Steering Committee Presentation [slides 698–705]
14:15–15:15	60 min	Hands-on: Lab 33: Capstone Presentation and Defence

		[slides 706–709] — capstone presentations to the steering committee panel
15:15–15:30	15 min	Tea break
15:30–15:45	15 min	Hands-on: Lab 34: Lessons Learned, Mentoring Green Belts and Your Black Belt Journey [slides 710–714] — lessons learned, mentoring Green Belts and your 90-day plan
15:45–16:00	15 min	Course recap and Briefing for Assessment [slides 715–726]
16:00–17:00	60 min	WSQ ASSESSMENT — Written Assessment (WA), Short-Answer Questions. 1 hour, open book
17:00–18:30	90 min	WSQ ASSESSMENT — Case Study (CS). 90 minutes, open book. PM digital attendance

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to the DMAIC phases)

DMAIC phase / Topic	Weighting	Labs
FOUNDATIONS: The Black Belt Role, Portfolio Selection & Your Capstone	10%	Lab 1, Lab 2, Lab 3
DEFINE: Define at Black Belt Depth — Charter, Stakeholders & Change	12%	Lab 4, Lab 5, Lab 6, Lab 7
MEASURE: Measure at Black Belt Depth — Advanced Measurement Systems	16%	Lab 8, Lab 9, Lab 10, Lab 11, Lab 12, Lab 13, Lab 14
ANALYZE: Analyze at Black Belt Depth — Advanced Statistical Analysis	22%	Lab 15, Lab 16, Lab 17, Lab 18, Lab 19, Lab 20, Lab 21
IMPROVE: Improve at Black Belt Depth — Design of Experiments	18%	Lab 22, Lab 23, Lab 24, Lab 25, Lab 26
CONTROL: Control at Black Belt Depth — Advanced SPC & Sustaining	12%	Lab 27, Lab 28, Lab 29, Lab 30
CAPSTONE: Capstone Project & Steering Committee Presentation	10%	Lab 31, Lab 32, Lab 33, Lab 34

Lab Data Files and Materials

Every lab folder under labs/ contains the worksheet (README.md) and the Excel workbook listed below. Datasets carry simulated process data for analysis; templates are blank structured forms the learner completes. All workbooks open on a Data Dictionary sheet defining each column, its unit and the specification limits. Learners are encouraged to substitute their own workplace data wherever they can obtain it.

Lab	Workbook	Type	What it contains
Lab 1	lab-01-project-selection-and-charter.xlsx	Template	A project selection matrix pre-loaded with six candidate projects, and a blank charter form.
Lab 2	lab-02-portfolio-and-belt-roles.xlsx	Template	A portfolio of eight live projects to balance against strategy, plus a belt role matrix.
Lab 3	lab-03-baseline-sigma-calculator.xlsx	Template	Twelve months of production and defect counts — compute DPU, DPO, DPMO, yield, RTY and the baseline sigma level for your project Y.
Lab 4	lab-04-voc-affinity-kano.xlsx	Template	Forty raw customer verbatims from hospital distributors and biomedical engineers — cluster them, translate them into requirements and classify them on the Kano model.
Lab 5	lab-05-ctq-tree-flowdown.xlsx	Template	A CTQ tree worksheet that flows the must-be need down to measurable, controllable Xs.
Lab 6	lab-06-sipoc-process-map.xlsx	Template	A SIPOC frame and a swimlane step list for the seal weld process, with handoff analysis.
Lab 7	lab-07-stakeholder-analysis.xlsx	Template	Twelve named stakeholders on this project — map their influence and support, and plan the specific action that moves each one.
Lab 8	lab-08-value-stream-map.xlsx	Template	Timed process data for the current-state value stream map, with the eight wastes observation log and a takt time calculator.
Lab 9	lab-09-data-collection-raw-extract.xlsx	Dataset	Raw 240-row MES extract of seal weld burst-pressure tests — deliberately messy, so you can write operational definitions that fix it.
Lab 10	lab-10-sampling-frame.xlsx	Dataset	A 600-unit population frame for one month — draw random, stratified,

			systematic and cluster samples from it and compare what each one tells you.
Lab 11	lab-11-gage-rr-continuous.xlsx	Dataset	Crossed Gage R&R: 10 parts x 3 operators x 3 trials on the burst-pressure rig. The study is built to FAIL — find out why before you trust any other data.
Lab 12	lab-12-attribute-msa-kappa.xlsx	Dataset	Attribute agreement study — 50 seal-weld photos, 3 inspectors, 2 trials each, against a known expert reference standard.
Lab 13	lab-13-nested-destructive-gauge.xlsx	Dataset	Nested (hierarchical) MSA for a DESTRUCTIVE burst test — each specimen can only be measured once, so operators measure different specimens from the same batch.
Lab 14	lab-14-baseline-capability.xlsx	Dataset	250 baseline burst-pressure readings in 50 subgroups of 5 (normal), plus 200 right-skewed leak-test cycle times for the non-normal capability exercise.
Lab 15	lab-15-run-chart-pareto-stratification.xlsx	Dataset	Three sheets: 60 daily Y averages for run-chart pattern analysis, a defect-mode Pareto count, and a plant/line/shift stratification table.
Lab 16	lab-16-cause-prioritisation.xlsx	Template	The 12 candidate causes the team brainstormed onto the fishbone — score and rank them into the vital few worth testing statistically.
Lab 17	lab-17-hypothesis-testing.xlsx	Dataset	Three tests on one sheet each: a 2-sample t (old vs new fixture), a paired t (before/after operator re-training) and a 2-proportion test (defect rates by plant).
Lab 18	lab-18-multiple-regression.xlsx	Dataset	120 production runs with five candidate Xs against burst pressure — including two predictors that are collinear on purpose.
Lab 19	lab-19-anova-interactions-nonparametric.xlsx	Dataset	One-way ANOVA across four seal-ring suppliers, a two-way ANOVA (plant x shift) containing a genuine interaction, and skewed data for Kruskal-

			Wallis.
Lab 20	lab-20-multivari-chisquare.xlsx	Dataset	A multi-vari study sampling positional, cyclical and temporal families, plus a plant x defect-mode contingency table for chi-square.
Lab 21	lab-21-analyze-tollgate-evidence.xlsx	Template	The evidence register for the Analyze tollgate — every claimed root cause, the statistical test that proves it, and the p-value.
Lab 22	lab-22-solution-selection.xlsx	Template	Candidate solutions generated against the proven root causes — screen and score them into an implementation set.
Lab 23	lab-23-full-factorial-doe.xlsx	Dataset	A replicated 2^3 full factorial (16 runs) on clamp pressure, dwell time and ambient temperature — run order already randomised.
Lab 24	lab-24-fractional-factorial-doe.xlsx	Dataset	A $2^{(5-1)}$ resolution V fractional factorial — 16 runs instead of 32, five factors, generator E = ABCD.
Lab 25	lab-25-rsm-central-composite.xlsx	Dataset	A 13-run central composite design (4 factorial + 4 axial + 5 centre points) around the Lab 23 optimum, to find the true peak and test for curvature.
Lab 26	lab-26-fmea-worksheet.xlsx	Template	A process FMEA worksheet pre-loaded with the seven real failure modes from this project — you score Severity, Occurrence and Detection and compute the RPN.
Lab 27	lab-27-spc-control-charts.xlsx	Dataset	Post-improvement control data: 30 subgroups of 5 for an Xbar-R chart, 40 individual clamp-pressure readings for an I-MR chart, and 24 weeks of defective counts for a p chart.
Lab 28	lab-28-cusum-ewma-shortrun.xlsx	Dataset	60 individual readings containing a small 0.75-sigma sustained shift at point 31 — plus a short-run sheet with three part numbers and only 15 units each.
Lab 29	lab-29-multivariate-spc-control-plan.xlsx	Dataset	50 paired clamp/dwell readings where the two variables are normally correlated — plus the project's draft control plan

			to govern.
Lab 30	lab-30-benefit-validation.xlsx	Dataset	Before/after cost data, implementation costs and the project's headline metrics — validate the benefit the way Finance will.
Lab 31	lab-31-a3-storyboard.xlsx	Template	The A3 storyboard skeleton and an artifact traceability register that maps every lab output onto its panel.
Lab 32	lab-32-steering-committee-deck.xlsx	Template	A slide-by-slide plan for the 15-minute steering committee presentation, plus the anticipated challenge log.
Lab 33	lab-33-capstone-defence-rubric.xlsx	Template	The rubric your capstone is assessed against, plus a peer-review form for the labs you observe.
Lab 34	lab-34-lessons-learned-mentoring.xlsx	Template	A lessons-learned register, a Green Belt mentoring plan and your personal Black Belt development plan.